

Efficient In-Context Learning for 3D Medical Image Segmentation

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by Tidiane Camaret Ndir



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Efficient In-Context Learning for 3D Medical Image Segmentation

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Declaration

I hereby declare, that I am the sole author and composer of my thesis and that no other sources or learning aids, other than those listed, have been used.

Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.

I also hereby declare that my thesis has not been prepared for another examination or assignment, either in its entirety or excerpts thereof.

Freiburg, September 27, 2026

Place, date

Tidiane Camaret Ndir

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Abstract

Zusammenfassung

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1 Motivation

In-context learning (ICL) recasts segmentation as few-shot matching: given a target scan and a small set of (image, mask) context pairs illustrating the structure of interest, a model predicts a mask for the target with no task-specific fine-tuning. This removes the need to retrain for every new anatomical class – valuable for structures that are rare, newly defined, or annotated at only a handful of sites.

Most ICL segmentation models were designed and validated in 2D. Moving to native 3D volumes is attractive – CT/MRI are inherently volumetric, and slice-wise processing discards through-plane context needed to resolve thin or elongated structures – but it sharpens an already difficult resolution/compute trade-off: dense cross-attention between a target volume and its context volumes scales with the cube of side length, so a design affordable at 64^3 becomes impractical at 256^3 .

This trade-off is not one problem but three, and this thesis addresses each with its own mechanism:

1. **How should image and label evidence be combined?** Concatenating image and mask channels before a shared encoder, or fusing them once through a single operation, commits to one fusion point before most of the network’s capacity is applied. section 3 proposes *bi-axial image–label attention*, keeping the two streams separate and letting them exchange information repeatedly.
2. **Where should compute be spent?** Re-scanning an entire volume with a sliding window at every processing step wastes compute on regions already known not to contain the structure of interest. section 3 proposes a *coarse-to-fine cascade* that restricts fine processing to the region a coarse pass has already localized, connected across levels by a query prior and register tokens that carry a gradient path between resolutions.
3. **Where does training signal come from beyond limited annotation?** Real (image, mask) pairs are limited to the classes and appearance distribution covered by the training cohort. section 3 proposes a *synthetic task generator* that repaints real anatomical geometry with novel, controllable appearance, producing label-perfect training tasks beyond the annotated cohort.

Contributions.

- A bi-axial image–label attention mechanism for in-context 3D segmentation, contrasted against early-concatenation (Medverse [7]) and one-shot fusion (Iris [5]) baselines.
- A coarse-to-fine cascade in which the query prior, region-restricted fine processing, and cross-level register training are evaluated against the sliding-window-over-the-whole-volume design used by prior work.
- A GMM-based synthetic task generator with a texture-noise variant that better matches real tissue’s spatial autocorrelation, evaluated against real-only and i.i.d.-noise training.
- Evaluation on TotalSegmentator CT and a range of out-of-distribution cohorts and modalities, reporting where each mechanism helps and where it does not.

Outline. section 2 introduces in-context segmentation, the volumetric encoder and attention machinery the method builds on, and synthetic-data background, then positions UniverSeg, Iris, and Medverse against the three axes above. section 3 details the model. section 4 evaluates each axis in turn. section 5 synthesizes what the evidence supports and what remains open, and section 6 summarizes.

2 Fundamentals

2.1 In-Context Segmentation

Given a target image I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same class, in-context segmentation predicts a mask $\hat{L}^t$ for the target in a single forward pass, with no gradient update and no class-specific parameters. This sits between two more familiar paradigms: closed-set supervised segmentation, which fixes a label set at training time and must be retrained (or fine-tuned) to add a class; and interactive segmentation, where a human supplies clicks or boxes on the target image itself, per image, at inference time. In-context segmentation instead supplies the “prompt” as a handful of *other*, already-labeled examples of the same class – closer to how a radiologist might be shown a few example cases of a rare finding and asked to find it in a new scan. This is valuable in medical imaging specifically because annotation requires domain expertise, new structures and pathologies are defined faster than segmentation datasets can be built for them, and per-site protocol differences limit how far a closed-set model trained elsewhere transfers.

2.2 Volumetric Encoders and Attention

Convolutional volumetric backbones. 3D U-Net [4] and its self-configuring descendant nnU-Net [8] encode a volume through a downsampling encoder into a feature pyramid, then decode back to full resolution via a mirrored upsampling path with skip connections carrying fine detail past the bottleneck. Deeper encoder stages have coarser spatial extent and more channels; shallower stages retain spatial detail at low channel count. This split – a coarse, semantically rich bottleneck feeding a computationally light attention stack, and fine, spatially precise early stages reserved for the decoder – is exactly what the coarse/fine tap split in subsection 3.1 reuses. UNETR [6] is a prior example of combining a convolutional decoder with transformer attention for 3D segmentation.

Attention. Scaled dot-product attention [13] computes, for each query token, a weighted average of value vectors, with weights given by the similarity between the query and each key:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \quad (1)$$

Multi-head attention runs several such operations in parallel subspaces and concatenates the results, letting different heads attend to different relationships in the same input.

Position encoding. Attention itself is permutation-invariant, so position must be injected separately. Rotary position embedding (RoPE) [11] does this by rotating each query/key vector by an angle proportional to its position before the dot product, so the result depends only on the *relative* position between the two tokens rather than their absolute indices. In 3D, this is applied axially: the head dimension is split into three chunks, one per spatial axis, each rotated by an angle proportional to that axis’s physical coordinate (scaled by voxel spacing) – letting attention between two cells at different cascade resolutions still reflect their true physical distance, not just their grid index.

2.3 Synthetic Training Data

Domain randomization trains a model on procedurally varied synthetic data so it does not overfit to the narrow appearance distribution of a single acquisition protocol or training cohort. SynthSeg [1] demonstrated this at scale for medical segmentation: training pairs are generated by sampling a random Gaussian-mixture appearance conditioned only on a label map, so the resulting segmentation network never sees a real image of any specific contrast during training yet generalizes across MRI contrasts and resolutions. GIN/IPA [9] takes a complementary, appearance-only approach to single-source domain generalization: a random nonlinear intensity warp (GIN), optionally blended across spatially distinct regions of the same volume (IPA) to break spurious global appearance correlations, achieving strong generalization from randomizing appearance alone, without touching shape or geometry. Both results motivate treating appearance randomization as a distinct lever from geometric augmentation – a question the synthetic task generator (subsection 3.4) revisits directly by ablating *how* painted appearance is randomized (i.i.d. vs. spatially correlated texture noise) while holding geometry fixed.

2.4 Related Work

Three prior systems attempt in-context medical image segmentation, each making a different choice on the three axes this thesis organizes around.

UniverSeg [2] popularized the approach with a CrossBlock architecture that densely cross-attends target and context features at full resolution – no coarse-to-fine structure, and no synthetic-task training.

Iris [5] encodes each context pair into a compact per-class task embedding and fuses it into the target through a pixel shuffle-unshuffle operation – a single fusion step, applied once, with no iterative refinement across scales.

Table 1: Positioning against prior in-context segmentation models on the three axes this thesis organizes around.

	Fusion	Coarse-to-fine	Synthetic tasks
UniverSeg	dense cross-attn.	no	no
Iris	one-shot pixel shuffle	no	no
Medverse	early concatenation	full-volume re-scan	no
Ours	bi-axial (repeated)	region-restricted, register-carried	GMM repaint + texture

Medverse [7] performs next-scale autoregressive segmentation, inspired by next-scale image generation [12]: at each step, the target and the previous step’s (perturbed) prediction are concatenated and processed by a shared U-Net, together with the real context pairs, correlated via a blockwise cross-attention module that pools queries and keys to block resolution for efficiency (subsection 3.2). Every step still re-scans the *entire* volume with a sliding window – only the input changes with scale, not the spatial extent processed.

Deep Neural Patchworks [10] stacks levels of increasing resolution for large-volume segmentation, but targets single-model supervised segmentation rather than in-context, few-shot prediction – an architectural precedent for coarse-to-fine cascades outside the ICL setting.

3 Methodology

3.1 Problem Setup and Backbone

Given a target volume I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same anatomical class as the (unknown) target mask, the model predicts a target mask $\hat{L}^t$. Prediction proceeds through a coarse-to-fine cascade of M resolution levels with decreasing voxel spacing $r_1 > r_2 > \dots > r_M$; each level produces a full mask that seeds the next, finer level (subsection 3.3). Figure 1 shows our architecture (Medverse’s, for comparison, is shown in Figure 4 in the appendix).

Shared encoder. All $K + 1$ volumes of a task are passed once through a shared encoder, producing a feature pyramid per volume. Two disjoint subsets of its stages are read out: *coarse* stages, resampled to a common $R \times R \times R$ token grid ($N = R^3$ cells per volume) and consumed by the attention stack below; and *fine* stages, kept at native resolution and reserved for the decoder’s skip connections (subsection 3.2). This split keeps the attention stack on a small, fixed-size token grid while still allowing a full-resolution decoded mask.

Tokenization. Each cell of every volume yields two tokens sharing one row – an image token z^{img} and a label token z^{lbl} – assembled into a single sequence together with a small number of learned *register tokens*, shared across all volumes, that give the attention stack fixed-size read/write capacity not tied to any single spatial cell. Context rows carry the ground-truth mask in their label token; the target’s does not, since that is exactly what is being predicted – it instead starts from a neutral placeholder or, at cascade levels beyond the first, the previous level’s own prediction (subsection 3.3).

3.2 Bi-Axial Image–Label Fusion

How a model combines image and label evidence, and how context and target interact, are two related design choices this section addresses together. Medverse [7] concatenates image and label channels before a shared U-Net; context and target then interact through a Blockwise Cross-Attention Module (BAM) placed at two points in that network (once encoder-side, once decoder-side), which pools its queries and keys to block resolution before attending – trading positional precision for a cheaper attention cost – while its values stay at full resolution. Iris [5] fuses image and label features once, through a pixel shuffle–unshuffle operation, then lets context inform the target through a single bidirectional cross-attention step against a compressed, per-class task embedding computed once and reused across every query. Both commit every design choice – fusion point and context–target interaction alike – to a single, early step, computed once.

Draft diagram, ported from the conference draft; finalize labels/scale once the architecture is settled.

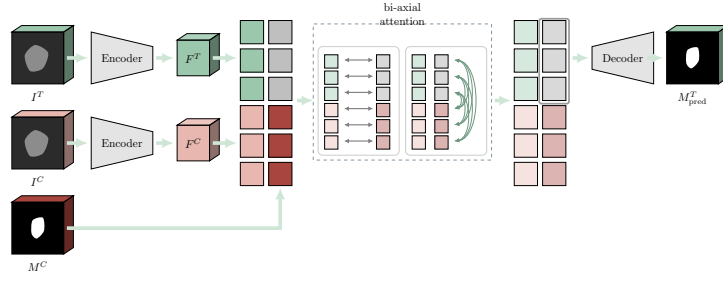


Figure 1: Architecture overview: separate image/label streams + bi-axial attention.

We instead keep image tokens z^{img} and label tokens z^{lbl} (subsection 3.1) as separate streams throughout the network and let them exchange information repeatedly, over two attention axes applied at every layer, on the full, uncompressed token grid:

- **Slot axis:** at each cell, the image and label token attend to each other, letting label evidence sharpen the image representation and vice versa, independently per cell.
- **Row axis:** every cell of every volume – context, target, and the register tokens – attends over the entire token sequence at once, using 3D axial rotary position embeddings so attention is a function of physical distance between cells rather than raw grid index. This single joint attention performs within-volume self-attention and cross-context matching together, since tokens from every volume already share one sequence.

Figure 2 illustrates both axes within a single layer.

Only the target’s post-attention *image* token is read out for decoding; the label token is discarded after the last layer, since it started from a placeholder rather than real image evidence for the target row. The decoded image tokens are projected back to the coarse token grid and progressively fused with the encoder’s fine-resolution skip features (subsection 3.1) via upsampling and convolution, one step per fine stage, into the final full-resolution mask logits.

3.3 Coarse-to-Fine Cascade

The cascade connects consecutive resolution levels through three mechanisms, each replacing a coarser design choice made by Medverse or Iris.

Query prior. Medverse [7] feeds an upsampled, perturbed ground-truth mask into an extra U-Net branch at every step; Iris [5] uses no prior at all. At level $\ell > 1$, we instead

Draft diagram, ported from the conference draft; finalize once the layer factorization is settled.

initialize the target’s label token directly from level $\ell-1$ ’s own prediction:

$$z_{\ell,0}^{\text{lbl}} = \text{LblEmbed}(\text{Warp}_{\ell-1 \rightarrow \ell}(\hat{L}_{\ell-1}^t)), \quad (2)$$

where Warp resamples level $\ell-1$ ’s prediction onto level ℓ ’s crop grid, correcting for the two levels’ different centers, field of view, and any augmentation applied independently to each. This makes the previous level’s belief part of the same token stream that attends bi-axially with the image (subsection 3.2), rather than a separate side channel. Figure 3 shows this flow across two levels.

Region-restricted fine processing. Medverse and Iris re-scan the entire volume with a sliding window at every step. Once a coarse level has localized the structure of interest, we instead restrict the sliding window at the next, finer level to the coarse prediction’s foreground region, thresholded and padded by a fixed margin. Compute at fine levels then scales with the size of the structure rather than the size of the input volume, at the cost of depending on the coarse level’s localization being approximately correct.

Cross-level training via register tokens. Medverse and Iris connect levels only through the rendered prediction feeding the next step, with no gradient path between them. The register tokens already introduced in subsection 3.1 give this a second, learned path: their final state $\mathbf{r}_\ell$ at level ℓ is carried into level $\ell+1$ via a learned projection and a level-independent type embedding:

$$\mathbf{r}_{\ell+1,0} = W_r \mathbf{r}_{\ell,\text{final}} + \mathbf{e}_{\ell+1}. \quad (3)$$

Unlike the query prior, this path is not detached between levels: level $\ell+1$ ’s loss can backpropagate into level ℓ ’s own attention, giving the model a direct incentive to carry forward whatever helps the next level – e.g. uncertainty or cues about neighboring structures – beyond what the rendered mask alone can encode.

3.4 Synthetic Task Generation

Real (image, mask) pairs are limited to the anatomical classes and appearance distribution covered by the training cohort. To broaden training beyond this, a fraction p_{synth} of training tasks are drawn from a synthetic generator instead of real context/target subjects.

Supervoxel repainting. A real-anatomy volume is first oversegmented into supervoxels (unsupervised, appearance-agnostic regions). One supervoxel is treated as the

Draft diagram, ported from the conference draft; finalize once the number of levels M is settled.

target class, and its binary mask is label-perfect by construction. Each of the $K+1$ volumes needed for a task is then produced by repainting the same supervoxel geometry with independently sampled per-class Gaussian intensities (a Gaussian mixture over classes), giving $K+1$ (image, mask) pairs that share identical anatomy but differ in appearance – no real cross-subject registration or annotation is required.

Paint noise. The default painting noise is i.i.d. per voxel: it matches each class’s target intensity variance but has near-zero spatial autocorrelation, unlike real CT/MRI tissue (measured ≈ 0.6 autocorrelation at native resolution, versus ≈ 0.0 for i.i.d. paint). An optional multi-octave variant instead sums several coarse-to-fine random fields before painting, producing spatially correlated texture that better matches real tissue statistics while leaving the per-class intensity variance unchanged.

3.5 Training Objective

Each cascade level is supervised independently at its own resolution with a combined binary cross-entropy and soft Dice loss, summed over the M levels:

$$\mathcal{L} = \sum_{\ell=1}^M \lambda_{\ell} (\mathcal{L}_{\text{BCE}}^{\ell} + \mathcal{L}_{\text{Dice}}^{\ell}) . \quad (4)$$

Level weights λ_{ℓ} are uniform ($\lambda_{\ell}=1$) in the reference configuration; the query-prior path is detached between levels (subsection 3.3), so each level’s loss trains its own prediction without back-propagating through the coarser level’s mask.

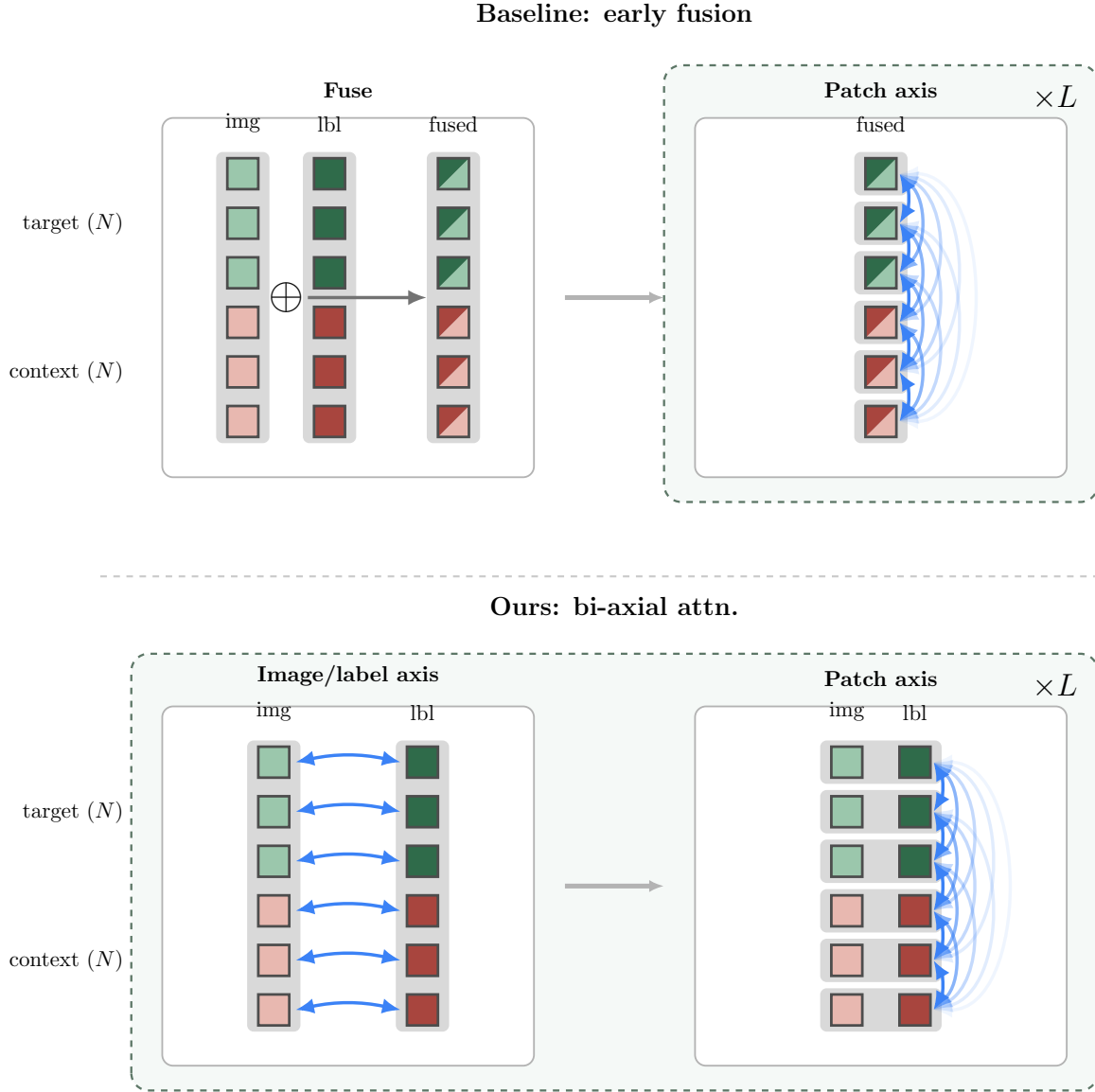


Figure 2: Bi-axial attention within one transformer layer, shown for a single context volume ($K=1$ for clarity) and 3 of its N spatial tokens per stream. Left: the slot axis – image and label tokens attend to each other within the same cell. Right: the row axis – tokens attend across cells and, for the target, across contexts. Green edges touch the target row, which cross-attends read-only into context and whose label-stream input is the query prior (subsection 3.3) rather than ground truth.

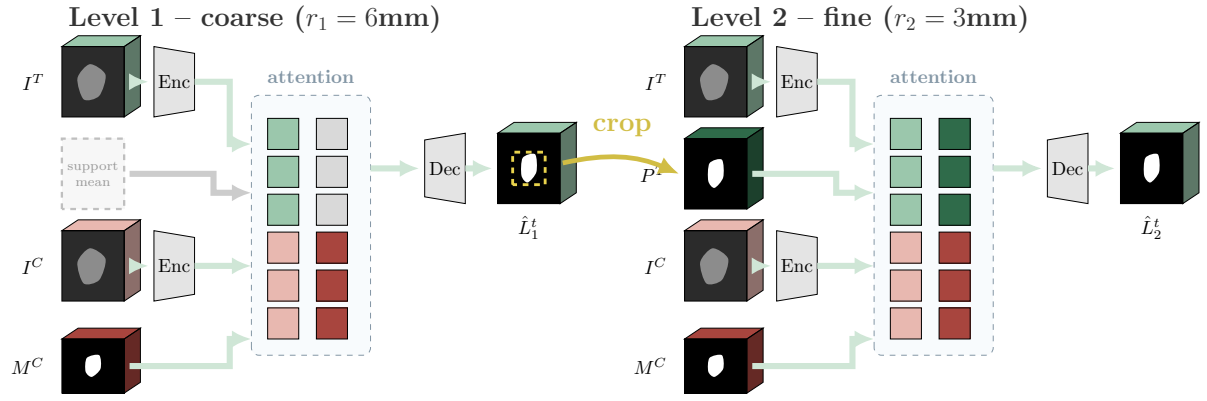


Figure 3: Query image prior across two cascade levels, with the full model input/output flow. At level 1 the target has no real prior yet (dashed support-mean placeholder); at level 2 the target's second lane is a real crop of level 1's own prediction $\hat{L}_1^t$ – the same lane the context mask always occupies.

4 Results

4.1 Experimental Setup

Model scope. Two configurations of the architecture are reported in this chapter, kept explicitly separate. The reference configuration (bi-axial attention, query-prior init, register carry gated off) is the source of every Fusion and Cascade number below. An extended configuration (adding cascade registers, a pooling token, and synthetic texture noise on top of it) is the source of every Synthetic Task Generation number. The two are trained independently and not directly comparable to each other – the extended configuration is read as a promising newer result, not a replacement for the numbers reported above it.

Baselines. Medverse [7] (released weights, and – where noted – finetuned on the same TotalSegmentator split) is the only baseline evaluated empirically here.

Metrics. Accuracy is reported as Dice (macro, per-class then averaged, unless stated per-sample) and Normalized Surface Dice (NSD). Compute is reported as GFLOPs, wall-clock latency, and peak VRAM where available; VRAM comes from a standalone $B=1, K=1$ inference benchmark rather than the evaluation loop itself, so it should not be read as representative of a real $K>1$ eval run.

Datasets. In-distribution evaluation uses TotalSegmentator [14] (117 classes, CT). Generalization is tested on the held-out sources of Table 2, spanning CT and MRI; most introduce classes entirely outside the TotalSegmentator vocabulary, but not all – FLARE22’s organs are a subset of the same vocabulary under a different dataset, so its classes split into a genuine seen/unseen pair the same way TotalSegmentator’s own held-out classes do (subsection 4.2).

*Sourced from an unofficial data mirror; provenance not yet independently confirmed.

A ninth check does not fit this table at all: TotalSegmentator’s own subjects also ship an MRI acquisition, evaluated in subsection 4.2 to isolate a held-out *modality* from a held-out *dataset* or *vocabulary* – same classes, same subject population, same institution, only the scanner modality is absent from training.

4.2 Image–Label Fusion

Even a conservative fusion design beats the field’s baseline, across held-out sources. Table 1 classifies Medverse’s fusion strategy as early concatenation: image

Resolve before submission: does the thesis report a single “Ours” configuration, or keep this split?

No Iris baseline exists yet; the comparison in subsection 2.4 remains architectural, not empirical.

Fill in n for the sources left

Table 2: Held-out generalization sources. Most introduce classes and/or modalities absent from training; FLARE22 is the exception, reusing TotalSegmentator’s own vocabulary under a different dataset.

Source	Modality	Target	n
ISLES22	MRI DWI	stroke lesion	–
Shifts-MS	MRI FLAIR	MS lesion	46
MSD Hippocampus	MRI T1	hippocampus	–
MSD Prostate	MRI T2+ADC	prostate zone	–
ATLAS v2.0*	MRI T1	chronic stroke lesion	–
GNC_705	MRI Dixon	kidney lesion	–
HU_LWK1	CT	L1 vertebra ROI	36
FLARE22	CT	13 abdominal organs	–

and label channels are merged before a shared U-Net processes them, committing to a single fusion point at the very start of the network. This section asks whether that architectural difference, independent of every other design choice this thesis makes elsewhere, already produces a measurable difference in practice – and does so at the most conservative point of comparison available, and across every dataset this thesis evaluates (subsection 4.1), not TotalSegmentator alone. The single-axis control introduced below collapses image and label evidence into one token per cell via a pixel shuffle–unshuffle operation *before* the transformer; of every variant of our architecture, it is structurally the closest to Medverse’s own commitment to early fusion. Comparing Medverse against this weaker control, rather than against the full bi-axial model, is the more conservative claim: if even the conservative variant wins, the win cannot be attributed to the slot axis alone. Medverse (released weights) and the single-axis control are trained under an identical data pipeline, augmentation recipe, and optimizer schedule, and evaluated with an identical protocol – fixed 3 mm single-level crops, fp32, each model’s own best checkpoint (no epoch matching across arms; Medverse trained to full convergence, the patchset arms did not).

Rows below are grouped by each dataset’s relationship to the training vocabulary, not by how different the source otherwise looks: TotalSegmentator itself and FLARE22 both reuse the training vocabulary and so split into a genuine seen/unseen pair (subsection 4.1); MSD Prostate and MSD Hippocampus introduce organs with no vocabulary overlap at all, so every class in that pair is unseen by construction, not by a train/val split; HU_LWK1’s target is a HU-measurement region, not a segmentable anatomical object, a categorically different kind of “unseen” from a novel organ; TotalSegmentator’s own MRI acquisition isolates a held-out *modality* instead, over classes the vocabulary already contains.

The single-axis control wins the in-vocabulary group outright – both TotalSegmentator

Table 3: Medverse (released baseline) vs. the single-axis control, across all evaluation datasets (subsection 4.1). Grouped by vocabulary relationship: in-vocabulary seen/unseen, novel-vocabulary (unseen by construction), non-object, held-out modality.

	Medverse (baseline)	Single-axis (ours, conservative)	n_{cls}
TotalSeg CT – seen	0.348	0.574	61
TotalSeg CT – unseen	0.221	0.408	56
FLARE22 – seen	0.532	0.693	11
FLARE22 – unseen	0.443	0.632	2
MSD Prostate (unseen)	0.377	0.205	2
MSD Hippocampus (unseen)	0.172	0.184	2
HU_LWK1 (non-object)	0.146	0.079	1
TotalSeg MRI (modality)	0.178	0.232	23
mean of 5 held-out sources	0.278	0.264	–
params	71.1M	74.1M	–
GFLOPs	2362.6	3973.1	–
latency (fp32, per sample)	73.2 ms	73.0 ms	–

splits and both FLARE22 splits – and also wins the held-out-modality case. Medverse wins MSD Prostate and the non-object case (HU_LWK1); the two are close on MSD Hippocampus. Averaged across the five held-out sources without weighting by class count, Medverse edges ahead (0.278 vs. 0.264) despite losing on more rows than it wins – generalization here is source-dependent, not a single uniform gap. The two novel-vocabulary and the non-object rows rest on very few classes (1–2 each) and should be read as directional, not conclusive. The compute picture is the honest opposite of a “cheaper” claim: the single-axis control’s accuracy advantage where it holds costs roughly 68% more GFLOPs, with parameter count roughly matched (+4%) and latency essentially tied once both are measured at the same precision. Two caveats keep this from a single-variable ablation: Medverse retains its released initialization rather than the randomly-initialized reasoning core both patchset arms share, and does not receive the same query-conditioning signal the reference model does during training, so this is best read as “released baseline vs. our current recipe.”

Isolating the slot-axis effect, across the same sources. Having shown that even the conservative variant is competitive with an external baseline, this ablation asks a narrower, fully controlled question: does keeping image and label as separate, repeatedly cross-attending streams (the slot axis, subsection 3.2) buy anything *on top of* an architecture that already competes, once every other confound is removed? The reference model (`arch.dual_axis=true`) is compared against the single-axis control from above (`arch.dual_axis=false`), which removes only the slot axis; the row axis

Re-read once Medverse receives the same query-conditioning signal – currently dropped due to a

Table 4: Bi-axial attention vs. the single-axis (early-fusion-style) control, at matched compute, across all evaluation datasets. Same grouping as Table 3.

	Single-axis (control)	Bi-axial (ours)	n_{cls}
TotalSeg CT – seen	0.574	0.582	61
TotalSeg CT – unseen	0.408	0.426	56
FLARE22 – seen	0.693	0.709	11
FLARE22 – unseen	0.632	0.604	2
MSD Prostate (unseen)	0.205	0.247	2
MSD Hippocampus (unseen)	0.122	0.184	2
HU_LWK1 (non-object)	0.079	0.102	1
TotalSeg MRI (modality)	0.232	0.259	23
mean of 5 held-out sources	0.264	0.297	–
l	9	4	–
params	74.1M	48.7M	–
GFLOPs	3973.1	3892.5	–
latency (fp32, per sample)	73.0 ms	72.5 ms	–

(cross-context attention) is identical in both.

Two adjustments relative to a naive dual-axis-flag toggle: (1) the single-axis control’s transformer depth is raised from $l=4$ to $l=9$ so its transformer compute is within 3% of the reference model’s – removing the slot axis also removes a large share of the model’s compute, and parameter count and compute could not be matched simultaneously by this route (matching compute instead leaves parameters 52% higher for the single-axis control); (2) both runs warm-start *only* the encoder and decoder from a shared checkpoint, with the entire in-context reasoning core (transformer, image/label embeddings, context/query id, register tokens, cascade projection, pool projection) randomly initialized for both arms, so neither model starts from an already-trained transformer. Both runs use real anatomy only ($p_{\text{synth}}=0$), CT only at train time, and the same evaluation protocol and dataset grouping as the table above.

Bi-axial attention wins every row except one: FLARE22’s two unseen classes, where single-axis is ahead (0.632 vs. 0.604) on a two-class sample too small to weigh heavily. Everywhere else – both TotalSegmentator splits, FLARE22’s seen classes, both novel-vocabulary datasets, the non-object case, and the held-out modality – bi-axial wins, and it wins the five-source mean (0.297 vs. 0.264) despite the one exception. With every other confound this thesis can control for removed, this is the cleanest, most consistently generalizing result in the axis. Latency is close to tied once both are measured at the same (fp32) precision, unlike an earlier bf16 measurement of the same pair that showed a larger gap – that gap looks like a precision/compilation artifact rather than a real cost

Table 5: Coarse-to-fine dice by cascade level, exp92 checkpoint, 1157 samples, spacing ladder [6, 3, 1.5]mm.

Level	6mm (coarsest)	3mm	1.5mm (finest)
Dice	0.3889	0.5088	0.5716

of the slot axis, and is not reflected here.

Status. Neither patchset arm had reached its full training budget at the point these numbers were read, while Medverse had already converged – so Table 3 may understate Medverse’s eventual ceiling more than it understates either patchset arm’s. Both comparisons are N=1 seed per arm, and several rows above rest on very few classes (FLARE22’s unseen split, both MSD sources, HU_LWK1) – directional evidence, not settled effect sizes. The params mismatch (+52% for single-axis) also leaves a residual capacity confound in Table 4 even after compute-matching.

4.3 Coarse-to-Fine Cascade

This section asks four questions about the coarse-to-fine cascade introduced in subsection 3.3, in order: whether it helps in-distribution at all; which of its mechanisms are responsible; whether that benefit survives a shift in modality and class vocabulary; and whether the individual mechanisms do too.

Coarse-to-fine cascading improves accuracy in-distribution. A spacing sweep from 4mm to 1.5mm crop resolution (117 classes, 3897 samples) raises macro Dice from **0.361 to 0.538** (+0.177). The locator’s containment of the target structure is already high at the coarse level (0.922 mean, vs. an oracle ceiling of 0.970 – a 0.048 gap) and does not predict which classes gain the most at the fine level: once a class is roughly localized, the remaining error is dominated by segmentation quality at the fine crop, not by further localization failure. A second, independent measurement on the exp92 checkpoint confirms the same monotonic pattern across a full three-level cascade (Table 5).

Accuracy improves monotonically from coarsest to finest level, at a cost of 11673 GFLOPs and 666ms wall-clock for the full three-level pass.

Which mechanisms drive that gain. Having established that the cascade helps, this subsection isolates which of its design choices are responsible: a controlled ablation of two of the cascade mechanisms introduced in subsection 3.3, the query prior and the cross-level register tokens. A three-arm chain – 145 → 146 → 147 – resumes the

Re-read both tables once all three arms reach their full training budget; add ≥ 2 seeds before citing either effect size as settled.

Table 6: Query-prior source and register-carry ablation, TotalSegmentator macro Dice (119 classes), epoch 59.

Arm	Config	Dice	Δ
145	GT prior (perturbed), registers off	0.499	–
146	Predicted prior, registers off	0.525	+0.026
147	Predicted prior, registers on	0.505	–0.020

same starting checkpoint independently for each arm and changes exactly one setting per step, evaluated in-distribution on TotalSegmentator (119 classes, real anatomy only, $p_{\text{synth}}=0$) at a shared epoch-59 cutoff (60-epoch budget).

A real predicted prior beats a perturbed ground-truth prior. Arms 145 and 146 differ only in what feeds the query prior at the fine level: a perturbed ground-truth mask (dilated, eroded, shifted, and noised) versus the model’s own real previous-level prediction. The real prediction wins, $0.499 \rightarrow 0.525$ macro Dice, improving 90 of 119 classes (median $+0.026$). This is an ablation of *which* non-null prior source is better, not of whether a prior helps at all – subsection 4.3 does not yet contain an arm that drops the query prior entirely.

Register carry regresses accuracy. Arms 146 and 147 differ only in `cascade_registers`. Turning it on regresses macro Dice, $0.525 \rightarrow 0.505$, across 77 of 119 classes, and costs roughly 2.5% additional inference latency for it. The regression is larger on classes held out of training than on seen classes (-0.030 vs. -0.009 mean per-class Dice), and is not uniform across anatomy: the worst-hit classes are almost all repeated fine structures that co-occur many times in one volume – individual ribs and vertebral levels, with `brachiocephalic_vein_left` collapsing to Dice 0.0 at every evaluated epoch once registers are on. A smaller set of large, uniquely-shaped structures (lungs, skull, aorta) improve under registers, consistent with the carried state helping global context at the cost of confusing instance disambiguation among near-identical repeated structures.

Generalization is modality- and quality-gated, not free. The in-distribution gain above does not transfer unconditionally to held-out sources. Comparing single-level against cascaded prediction on the `exp92` checkpoint across three of the sources in Table 2 (Table 7) shows a consistent pattern: the cascade helps when the coarse level’s own single-level prediction is already reasonable and the modality matches training (HU_LWK1, CT), and hurts – badly, in one case – when the coarse level is already failing (ISLES22, GNC_705, both MRI, both lesion targets far outside TotalSegmentator’s anatomical-organ training distribution).

Add a no-query-prior control arm to complete the ablation table promised in the methodology chapter (full model vs. –query-prior vs. –region-restriction vs. –register-carry).

Table 7: Single-level vs. cascaded macro Dice, exp92 checkpoint, held-out sources.

Source	Modality	Single-level	Cascade	Verdict
HU_LWK1 (L1 vertebra)	CT	0.116	0.129 ([6, 3, 1])	helps
ISLES22	MRI DWI	0.054	0.046 ([3, 1.5])	hurts
GNC_705 (kidney lesion)	MRI Dixon	0.059	0.019 ([6, 3, 1.2])	hurts badly

Table 8: Same ablation as Table 6, evaluated out-of-distribution, macro Dice.

Source	n	145	146	147
HU_LWK1 (CT)	36	0.099	0.152	0.155
FLARE22 (CT)	650	0.731	0.751	0.744
NasalSeg (CT)	535	0.399	0.547	0.608
MSD Prostate	124	0.310	0.375	0.358
MSD Hippocampus	520	0.319	0.302	0.400
ISLES22	247	0.067	0.069	0.060
Shifts-MS	46	0.035	0.031	0.030
ATLAS v2.0	654	0.029	0.026	0.025
GNC_705	1490	0.027	0.024	0.021

An independent, indirect check on a different architecture corroborates the same mechanism: feeding Medverse’s own imperfect coarse prediction forward as a query prior hurts its cascade universally, across all 7 held-out sources tested, sometimes catastrophically (MSD Hippocampus $0.691 \rightarrow 0.050$). Taken together, this is evidence that compounding coarse-level error into the fine level is a property generic to coarse-to-fine cascades, not specific to this architecture’s design.

Neither in-distribution mechanism ablation replicates on held-out generalization sources either. The modality-gated pattern above is about whether cascading at all helps or hurts; this second check asks whether the specific mechanisms ablated above (prior source, register carry) keep the same direction OOD. The same three checkpoints, evaluated on nine held-out sources – the eight of Table 2 plus NasalSeg (CT, sinus/nasal-cavity classes, not yet added to that table), added 2026-09-26 (each source’s own cascade ladder; `msd_hippocampus` single-level only), give a mixed picture.

The prior-source change ($145 \rightarrow 146$) helps 5 of 9 sources and hurts 4 – close to an even split, notably weaker than the clear majority-positive direction seen in-distribution (90 of 119 classes). The register-carry change ($146 \rightarrow 147$) helps 3 of 9 and hurts 6 – a majority in the same direction as its in-distribution regression, though far from uniform: HU_LWK1’s cascade does not regress under registers here ($0.152 \rightarrow 0.155$), and two sources gain substantially instead – MSD Hippocampus (+0.097) and NasalSeg (+0.061). NasalSeg is the one source where both mechanism changes move in the same direction

Add NasalSeg to Table 2 in subsection 4.1 for consistency – it is evaluated here but

Table 9: Paint-noise ablation, TotalSegmentator validation Dice, epoch 50. Neither synthetic arm had converged at this cutoff (both still rising); the real-only arm had already plateaued.

Arm	Dice	seen	unseen
Real-only	0.4868	0.5737	0.3886
Flat i.i.d. noise	0.4899	0.5676	0.4022
Multi-octave texture	0.4955	0.5748	0.4059

as their in-distribution counterparts (prior-source +0.149, registers +0.061); FLARE22 only partially agrees, gaining from the prior-source change (+0.020) but losing slightly to register carry (−0.007).

4.4 Synthetic Task Generation

This section isolates the effect of the synthetic generator itself (subsection 3.4) – paint noise texture and shape/intensity calibration – from the model’s training data mix. All runs below are trained on CT only; the separate question of what adding MRI training data does to generalization is a training-regime choice, not a property of the synthetic generator, and is reported elsewhere rather than folded into these numbers.

Texture noise beats flat noise beats real-only. A controlled three-arm chain shares one starting checkpoint and varies only the paint-noise mechanism: real-only ($p_{\text{synth}}=0$), synthetic with i.i.d. paint noise ($p_{\text{synth}}=0.3$), and synthetic with multi-octave correlated texture noise (same p_{synth}). Metric: TotalSegmentator validation Dice, seen and unseen class macro splits, at a shared epoch-50 cutoff.

Texture noise wins on all three metrics – notably it recovers the seen-class cost that flat noise alone pays (narrowly beating even the real-only arm), while matching flat noise’s unseen-class gain.

Shape and intensity realism, calibrated against measurement. The procedural shape-mode generator’s host-anchored intensity contrast (subsection 3.4) was calibrated against a fresh measurement of real target-vs-surrounding-tissue contrast, taken directly on the same held-out sources used for generalization testing (749 cases, effect size in ring-standard-deviation units) rather than guessed.

Every source’s sign and rough magnitude matches its known radiological contrast direction, validating the measurement before it was used to set the generator’s calibration range. The measurement also caught a design error: an initial *absolute*-offset version

This table super-sedes an earlier, mid-training cascade_r OOD number for HU_LWK1 (0.1287 → 0.0935, from an earlier-checkpoint state of this same 145/146/14 chain) – do not cite that number. Not to be confused with Table 7’s 0.129,

Table 10: Measured target-vs-surrounding-tissue contrast, used to set the shape generator’s calibration range. Sign and magnitude match each source’s known imaging contrast pattern.

Source	Known pattern	Contrast (ring- σ)
ISLES22 (DWI, acute stroke)	hyperintense	+2.42
Shifts-MS (FLAIR, MS lesion)	hyperintense	+1.49
GNC_705 (kidney lesion)	mixed cyst/complex	+0.49
MSD Prostate (zones)	–	+0.35
HU_LWK1 (vertebra ROI, CT)	≈ 0 expected	–0.07
MSD Hippocampus	–	–0.51
ATLAS v2.0 (T1, chronic stroke)	hypointense	–0.59

of this calibration was, by this same measurement, roughly $7\text{--}12\times$ too extreme relative to the generator’s own per-class variance scale – corrected to a ratio-of-host-variance design before any reported checkpoint used it.

Not yet isolated from the training-mix change. Two further generator changes – cross-class intensity/variance correlation and widened shape diversity – exist and have been evaluated, but only on top of a checkpoint that concurrently switched the training data mix from CT-only to CT+MRI. Their standalone effect, isolated from that mix change, has not been measured.

Rerun the cross-class-correlation and shape-diversity ablations on a CT-only baseline (matching this section’s texture-noise arms) so the synthetic-generator effect

5 Discussion

6 Conclusion

ToDoS

Select the correct language in the first rows of `main.tex`. The % makes a line a comment.

Generate a PDF (Visual Studio Code: click on the green arrow in the top right corner).

Check the PDF (Visual Studio Code: click on the two rectangles with the magnifying glass in the top right corner).

Talk about this template with your supervisor: Should you add a declaration for the use of AI? Currently no official AI-declaration is given. You can find one in the pull requests of the github repository.

With CTRL pressed you can click on the PDF and jump to the corresponding place in your `.tex` files.

Check the Errors. (Visual Studio Code: bottom left) There should be one warning because the first ToDo pushes down the second one: LaTeX Marginpar on page 2 moved.

In the `main.tex` you can press ALT + SHIFT and the ARROW UP and DOWN keys to create a multiline cursor, comment out all the sections you are currently not working on and remove the multiline cursor with ESC.

Keep the following examples and todos till the end and check them out sometimes.

Adjust your License Notes to reflect the used programs.

For Printing: Remove the `todonotes`-package in `setup.tex` to find all Todos.

Table 11: A nice table with a nice caption.

Value & Unit	Value	Unit
9.82 m s ⁻²	9.82	m s ⁻²

For Printing: Choose the correct formatting (search `print` in `setup.tex`).

You can remove the glossary and the symbolslist to make it easier, especially in a Bachelor's Thesis.

6.1 Tables and References

Table 11

This is a cited book [3].

Store and display plots and other graphs as `.pdf` files.

Include Algorithm 1 in the same way you include equations like Equation 5:

$$\begin{aligned}y &= 3x^2 + 5x^2 + 4x + 4x + 7 + 1 \\&= 8x^2 + 8x + 8 \\y &= 8 \cdot (x^2 + x + 1)\end{aligned}\tag{5}$$

6.2 Auto-complete in VSCode

There are also some auto-complete features: type `@` + letter to get the letters greek version.

Type `@v` + letter for the alternative form (like φ for ϕ).

You can also use shortcuts for environments like

$$BAL\tag{6}$$

Type BAL and select the correct option from the autocomplete. Maybe you have to try various ones. The B is the base command and AL is the abbreviation for align.

If you want an environment without an index, you can use BS as base command.

Same for changes to your font, like bold or italic: FIT gives *text*

6.3 ", Quotations and Code

Use " or file-names, file_names and code_snippets.

Algorithm 1: The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in setup.tex.

```
from dataclasses import dataclass
```

```
# https://cd.uni-freiburg.de/farben/
```

```
@dataclass
```

```
class MainColors:
```

```
    blue = "#344A9A"
```

```
    black = "#000000"
```

```
    white = "#FFFFFF"
```

```
@dataclass
```

```
class BackgroundColors:
```

```
    blue_100_percent = "#344A9A"
```

```
    blue_80_percent = "#5D6BAD"
```

```
    blue_60_percent = "#868DC2"
```

```
    blue_80_percent = "#AFB1D8"
```

```
    blue_20_percent = "#D7D8EC"
```

```
    dark_blue = "#00004a"
```

```
    sand_100_percent = "#f6f1e3"
```

```
    sand_50_percent = "#faf8f1"
```

```
@dataclass
```

```
class AdditionalColors:
```

```
    green_100_percent = "#00a082"
```

```
    green_80_percent = "#27b29b"
```

```
    green_60_percent = "#7bc6b4"
```

```
green_80_percent = "#afdace"  
green_20_percent = "#daede7"  
brown_100_percent = "#8f6b30"  
brown_80_percent = "#a58959"  
brown_60_percent = "#bca683"  
brown_80_percent = "#d2c4ac"  
brown_20_percent = "#e9e1d6"  
yellow_100_percent = "#ffe863"  
yellow_80_percent = "#ffed82"  
yellow_60_percent = "#fff1a1"  
yellow_80_percent = "#fff6c1"  
yellow_20_percent = "#fffae0"  
pink_100_percent = "#f5c2ed"  
pink_80_percent = "#f7cef1"  
pink_60_percent = "#f9daf4"  
pink_80_percent = "#fbe7f8"  
pink_20_percent = "#fdf3fb"
```

6.4 Clipboard Manager, Git and more

Clipboard Manager help with copying multiple different things at once. On Windows you can press WINDOWS + V to activate the history and choose older things you copied.

Use Git ([click here](#)).

6.5 Glossary

Click them, the page numbers in the glossaries are also clickable.

name one

name one

name one

long form of name two (short2)

short2

short2

a symbol

a symbol

a symbol

Big Letter at start: Name one

plural of name one

name one

Big Letter at start: Short2

short2

short2s

Big Letter at start: *A symbol*

a symbol

Symbols

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Glossary

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name one	description one	21, 23
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Appendix

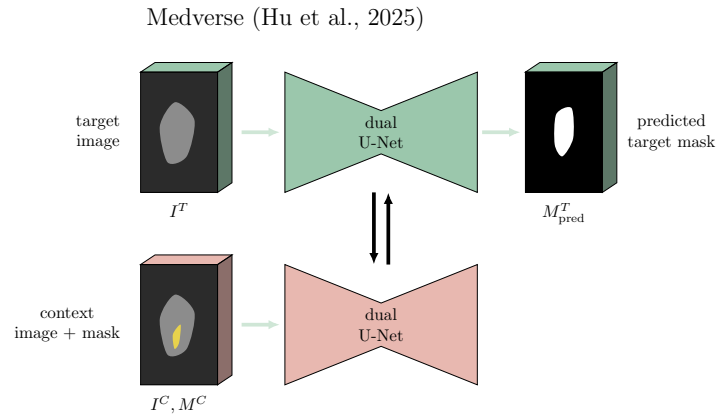


Figure 4: Medverse’s architecture: early concatenation + dual U-Net, shown for comparison against our architecture (Figure 1).